PRODUCTION LLM & AGENTIC AI ENGINEERING MASTER PROGRAM

Lead Instructor & CTO: Mian Ahmad Basit

Master 120+ in-depth topics across Custom LLM Fine-Tuning, Agentic Swarms, Computer Vision, AI Security, AIOps, LLMOps, RAG, agent protocols, and Kubernetes cloud deployment.

Program Length: 7-8 months | Modules: 24 | Access: 2 Years LMS Access

CURRICULUM DOMAINS

Phase 0 - Programming & AI Foundations: Python fundamentals, OOP, Async Concurrency, and project engineering prerequisites.

ML Phase - Machine Learning Foundations: Linear Algebra, Calculus, Statistics, Supervised and Unsupervised ML.

DL Phase - Deep Learning Foundations: Neural networks, backpropagation, PyTorch mechanics, CNNs, and Sequence Models.

NLP Phase - NLP Foundations: Text preprocessing, Word2Vec, GloVe, embeddings, and NLP downstream tasks.

Phase 1 - AI Foundations & Transformers 101: Transformer architecture, attention calculus, tokenization, positional encodings.

Phase 2 - Prompts to Programs & Programmatic LLMs: Prompt engineering, Pydantic schemas, Outlines grammars, DSPy optimization.

Phase 3 - KV Cache & Advanced Attentions: FlashAttention, MQA, GQA, MLA, PagedAttention, and scaling laws.

Phase 4 - Finetuning Language Models & PEFT: SFT, LoRA, QLoRA, DoRA, DPO, GRPO, ORPO, quantization, synthetic datasets.

Phase 5 - Mixture of Experts & Reasoning Models: MoE routing, DeepSeek-R1-Zero, Chain-of-Thought reinforcement learning.

Phase 6 - Knowledge Distillation & Small Language Models: SLM design, liquid models, KL divergence, student-teacher pipelines.

Phase 7 - Multimodal Vision Systems: ViT, CLIP, SigLIP, DINOv2, LLaVA, Qwen-2-VL.

Phase 8 - Speech AI: Whisper, STT foundations, acoustic datasets, and custom fine-tuning.

Phase 9 - Representation Learning: Dense, sparse, quantized, binary, ColBERT, MRL, embedding fine-tuning.

Phase 10 - Production RAG Systems: LangChain 2026, LlamaIndex, Hybrid RAG, Self-RAG, Corrective RAG, ColPali, GraphRAG, RAG Security.

Phase 11 - Agentic Systems: LangGraph, PydanticAI, Pydantic Evals, MCP, A2A protocols.

Phase 12 - Agent Orchestration: MCP, A2A, function calling, agent observability.

Phase 13 - Context Engineering: Context compression, LLMLingua, RECOMP, lm-eval, Inspect AI, E2B sandboxes.

Phase 14 - LLMOps & Observability: LangSmith, Langfuse, Logfire, RAGAS, caching, token budgeting.

Phase 15 - Production Infrastructure: vLLM, SGLang, llama.cpp, speculative decoding, Kubernetes, Docker, Helm, GitHub Actions.

Phase 16 - AI Security: NeMo Guardrails, LlamaFirewall, Presidio PII, LiteLLM gateways, RBAC.

OFFICIAL TOOLING COVERED

n8n, Claude 3.5 Sonnet, OpenAI GPT-4o & o3, NVIDIA NeMo & NIM, Hugging Face, DeepSeek-R1 & V3, Meta LLaMA 3.3, Google Gemini 1.5, LangGraph & LangChain, Qdrant & Pinecone, vLLM & Ollama, FastMCP & Docker/K8s.

CAREER OUTCOMES

Global AI Engineer Salary: $120K - $250K/yr.

Remote consulting rate: $80 - $250/hr.

Enterprise AI agency retainers: $5,000 - $25,000 per project.

SECURITY & COMPLIANCE

NVIDIA NeMo Guardrails, Microsoft Presidio PII masking, OWASP Top 10 for LLMs defense, private air-gapped VPC deployment, RBAC, prompt injection defense, and retrieval security.